

KHANG TRUONG GIANG

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Objective: Build autonomous systems to reconstruct and understand the 3D world from 2D images.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) <i>Ph.D. in Computer Science</i> Advisors: Prof. Sungho Jo, Prof. Soohwan Song. Research Interests: 3D Reconstruction, Image Matching, Visual Localization, Depth Estimation.	Daejeon, South Korea 09/2021 – 08/2024
Korea Advanced Institute of Science and Technology (KAIST) <i>M.S. in Computer Science</i>	Daejeon, South Korea 09/2019 – 08/2021
Hanoi University of Science and Technology (HUST) <i>B.Sc. in Computer Science (School of ICT Talent Program)</i> GPA: 3.62 / 4.0 Graduation with Honors (Excellent Degree)	Hanoi, Vietnam 09/2013 – 08/2018

PROFESSIONAL EXPERIENCE

42dot (Global Software Center of Hyundai Motor Group) <i>3D Vision Engineer</i>	South Korea 07/2024 – Now
• Autonomous Driving Perception: Architected and trained robust multi-camera perception models, including 2D/3D object detection, 3D traffic light detection, and diffusion-based trajectory planning. • Real-Road Deployment: Developed long-range models capable of detecting distant objects up to 150m and traffic light signals up to 90m, enabling end-to-end autonomous driving with minimal human intervention. • Data Engineering: Built automated and pseudo-labeling data pipelines to systematically generate large-scale datasets, accelerating perception model training cycles. • Peer Recognition: Voted by team members for exceptional contributions to the autonomous driving product and awarded a company-sponsored merit workcation to Jeju Island (2024).	
Neuro-Machine Augmented Intelligence Lab, KAIST <i>Research Assistant</i>	South Korea 09/2018 – 06/2024
• 3D Vision Research: Conducted cutting-edge research in core 3D reconstruction areas, including high-precision feature matching, visual localization, and multi-view depth estimation. • Framework Architecture: Authored, open-sourced, and actively maintained influential research repositories on GitHub, including: ◦ CDS-MVSNet: An accurate, state-of-the-art learning-based multi-view stereo network. ◦ TopicFM / TopicFM+: A highly robust, efficient, and interpretable image matching model. ◦ DeViLoc: A semi-dense visual localization system designed for robust spatial awareness. • Scientific Impact: Published multiple papers at premier AI/ML venues (CVPR, AAAI, ICLR, IJCAI, IEEE TIP).	
Viettel High Technology Industries Corporation <i>Software Engineer</i>	Hanoi, Vietnam 09/2018 – 06/2019
• Predictive Analytics: Built a robust machine learning pipeline to analyze and predict user activity levels across the telecommunications network. • API Deployment: Containerized and deployed the predictive models as production Web APIs using Flask.	

SELECTED PUBLICATIONS

[C4] B. Lee, J. Park, **Khang Truong***, and S. Song*. "Online 3D Gaussian Splatting Modeling with Novel View Selection." *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence (IJCAI)*, 2025. (**Top-tier in AI, *Corresponding Authors**)

[J5] B. Lee, J. Park, **Khang Truong**, S. Jo, and S. Song. "MVS-GS: High-Quality 3D Gaussian Splatting Mapping via Online Multi-View Stereo." *IEEE Access*, 2025. **(Impact Factor: 3.6)**

[C3] **Khang Truong**, S. Song, and S. Jo. "Learning to Produce Semi-dense Correspondences for Visual Localization." *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. **(Oral Presentation, top 3.3% of accepted papers / 90 out of 2,719)**

[J4] **Khang Truong**, S. Song, and S. Jo. "TopicFM+: Boosting Accuracy and Efficiency of Topic-Assisted Feature Matching." *IEEE Transactions on Image Processing (TIP)*, 2024. **(Impact Factor: 13.7)**

[C2] **Khang Truong**, S. Song, and S. Jo. "TopicFM: Robust and Interpretable Topic-assisted Feature Matching." *AAAI Conference on Artificial Intelligence (AAAI)*, 2023. **(Top-tier in AI, acceptance rate: 19.6% of 8,777 submissions)**

[J3] J. Mun, **Khang Truong**, Y. Lee, N. Oh, S. Huh, M. Kim, and S. Jo. "HybGrasp: A Hybrid Learning-to-Adapt Architecture for Efficient Robot Grasping." *IEEE Robotics and Automation Letters (RA-L)*, 2023. **(Impact Factor: 5.2)**

[J2] S. Song*, **Khang Truong***, D. Kim, and S. Jo. "Prior depth-based multi-view stereo network for online 3D model reconstruction." *Pattern Recognition*, 2023. **(Impact Factor: 8.0, *Co-first Author)**

[C1] **Khang Truong**, S. Song, and S. Jo. "Curvature-guided dynamic scale networks for multi-view stereo." *International Conference on Learning Representations (ICLR)*, 2022. **(Top-tier in ML/DL)**

[J1] **Khang Truong**, S. Song, D. Kim, and S. Choi. "Sequential Depth Completion With Confidence Estimation for 3D Model Reconstruction." *IEEE Robotics and Automation Letters (RA-L)*, 2020. **(Impact Factor: 5.2)**

HONORS AND AWARDS

Outstanding Reviewer Award, Conference on Computer Vision and Pattern Recognition (CVPR)	2024
Silver Medal (Top 9%), Kaggle Image Matching Challenge	2023
Hyundai Motor Chung Mong-Koo Foundation Global Scholarship for outstanding PhD students	2021 – 2024
Full Graduate Scholarship, KAIST Master's Program	2019 – 2021
Excellence Academic Scholarship, HUST	2018
Lotte Scholarship for outstanding university students	2017
Certificate of Excellence (Ranked 1st in academic cohort), HUST	2014
Third Prize, Vietnam Mathematical Olympiad (VMO) for High School Students	2013

ACADEMIC ACTIVITIES & SERVICE

Conferences:	CVPR, ICCV, AAAI, NeurIPS
Journals:	IEEE TIP (2024), IEEE TCSVT (2023), IEEE RA-L (2023), IEEE T-ASE (2024)
Mentorship:	Academic Mentor, advising two undergraduate and master's students in research initiatives.

COMPUTER SKILLS

Programming:	Python (Advanced), C/C++ (Basic)
Frameworks:	PyTorch (Advanced), PyTorch Lightning (Intermediate)
Miscellany:	Linux (Advanced), LaTeX, Git. Adaptive and swift learner of new technical tools/frameworks.